Lista - Integral - Matemática Elementar

1. Determine as integrais indefinidas:

1a. $\int \left(3t^2 + \frac{t}{2}\right) dt = F(t) = \frac{3t^{2+1}}{2+1} + \frac{t^{1+1}}{2\cdot1+1} = \frac{\cancel{3}t^3}{\cancel{3}} + \frac{t^2}{4} = t^3 + \frac{t^2}{4} + c$

1b. $\int (2x^3 - 5x^2 + 3x + 1)\,dx = F(x) = \int 2x^3\,dx - \int 5x^2\,dx + \int 3x\,dx + \int 1$

$= \frac{\cancel{2}x^{4}}{\cancel{4}} - \frac{5x^3}{3} + \frac{3x^2}{2} + x$

$= \frac{x^4}{2} - \frac{5x^3}{3} + \frac{3x^2}{2} + x + c$

1c. $\int \left(\frac{1}{x^2} - x^2 - \frac{1}{3}\right) dx = F(x) = \int \frac{1}{x^2}\,dx - \int x^2\,dx - \int \frac{1}{3}$

$= x^{-2} - \frac{x^3}{3} - \frac{1x}{3} + c$

$= \frac{-x^{-1}}{-1} - \frac{x^3}{3} - \frac{1x^3}{3} + c$

$= -x^{-1} - \frac{x^3}{3} - \frac{x}{3} + c$

1d. $\int (\sqrt{x} + \sqrt[3]{x})\,dx = F(x) = \int x^{\frac{1}{2}}\,dx + \int x^{\frac{1}{3}}$

$\frac{x^{\frac{1}{2}+1}}{\frac{1+1}{2}} + c + \frac{x^{\frac{1}{3}+1}}{\frac{1+4}{3}} + c$

$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{4}{3}}}{\frac{4}{3}}$

$= \frac{2\sqrt{x^3}}{3} + \frac{3\sqrt[3]{x^4}}{4} + c$

e. $\int \frac{x^2+1}{x}\,dx = F(x) \quad \int\left(\frac{x^{\cancel{2}}}{\cancel{x}} + \frac{1}{x}\right) = \int\left(x^1 + \frac{1}{x}\right) = \frac{x^2}{2} + \ln|x| + c$

f. $\int 5\sqrt[3]{x^2}\,dx = F(x) \quad 5\int x^{\frac{2}{3}}\,dx = \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1} = \frac{x^{\frac{5}{3}}}{\frac{5}{3}} = \frac{\cancel{5}\,3\sqrt[3]{x^5}}{\cancel{5}} + c$

g. $\int\left(8y - \frac{2}{y^{\frac{1}{4}}}\right)dy = F(y) \int 8y\,dy - \int \frac{2}{y^{\frac{1}{4}}}\,dy$

$= \frac{8y^{1+1}}{1+1} + c - \frac{2y^{-\frac{1}{4}+1}}{-\frac{1}{4}+1} + c$

$= \frac{8y^2}{\cancel{2}} - \frac{2y^{\frac{3}{4}}}{\frac{3}{4}} + c$

$= 4y^2 - \frac{4\cdot 2y^{\frac{3}{4}}}{4} + c$

$= 4y^2 - \frac{8y^{\frac{3}{4}}}{4} + c$

h. $\int 2x - 2x^{-2}\,dx = F(x) \int 2x\,dx - \int 2x^{-2}\,dx$

$= \frac{\cancel{2}x^2}{\cancel{2}} - \frac{2x^{-2+1}}{-2+1}$

$= x^2 - \frac{2x^{-1}}{-1}$

$= x^2 + 2x^{-1}$

$= x^2 + \frac{2}{x} + c$

i. $\int \left(\frac{1}{\sqrt{t}} + 4t + 1 \right) dt = F(t) \int \frac{1}{t^{\frac{1}{2}}} dt + \int 4t \, dt + \int 1 \, dt$

$= t^{-\frac{1}{2}} + 4t + 1$

$= \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + \frac{4t^2}{2} + t + c$

$= \frac{2t^{\frac{1}{2}}}{1} + 2t^2 + t + c$

$= 2\sqrt{t} + 2t^2 + t + c$

j. $\int (\cos(t) - \sin(t)) \, dt = F(t) \sin(t) + \cos(t) + c$

k. $\int 2e^t + \frac{1}{t} \, dt = 2e^t + \ln|t| + c$

2. Resolva os problemas de valor inicial

a. $\frac{dy}{dx} = 2x - 4, \; y(2) = 0$

$\int dy = \int (2x - 8) \, dx$

$y = \frac{2x^2}{2} - 4x + c$

$y = x^2 - 4x + c$

$y(2) = 2^2 - 4 \cdot 2 + c = 0$

$4 - 14 + c = 0$

$c = -4 + 14$

$c = 10$

$y = x^2 - 4x + 10$

b. $\frac{dy}{dx} = \frac{1}{x^2} + x \quad x > 0$ e $y(2) = 1$

$\int dy = \int \frac{1}{x^2} + x \rightarrow \int dy = \int x^{-2} + x$

$y = \frac{x^{-1}}{-1} + \frac{x^2}{2} + c$

$y = \frac{-1}{x} + \frac{x^2}{2} + c$

$y(2) = \frac{-1}{2} + \frac{2^2}{2} + c = 1$

$= -0{,}5 + 2 + c = 1$

$c = 1 + 0{,}5 - 2$

$c = 1{,}05 - 2$

$c = -0{,}5 = \frac{-1}{2}$

$y = \frac{-1}{x} + \frac{x^2}{2} - \frac{1}{2}$

c. $\frac{dy}{dx} = 3x^{-\frac{2}{3}} \quad y(-1) = -5$

$\int dy = \int 3x^{-\frac{2}{3}} dx$

$y = \frac{3x^{\frac{1}{3}}}{\frac{1}{3}} + c$

$y = 9x^{\frac{1}{3}} + c$

$y(-1) = 9(-1)^{\frac{1}{3}} + c = -5$

$9(-1) + c = -5$

$-9 + c = -5$

$c = 9 - 5$

$c = 4$

$y = 9x^{\frac{1}{3}} + 4$

Tudo que está no num lado que c muda de sinal

d. $\frac{dy}{dx} = 2e^x + 3x^2 \quad y(0) = 4$

$\int dy = \int 2e^x + 3x^2 dx$

$y = 2e^x + \frac{\cancel{3}x^3}{\cancel{3}}$

$y = 2e^x + x^3 + c$

$y(0) = 2e^0 + 0^3 + c = 4$

$2.1 + 0 + c = 4$

$2 + c = 4$

$c = 4 - 2$

$c = 2$

$y = 2e^x + x^3 + 2$

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e. $y'' = 2 - 6x$ $\quad y'(0) = 4 \quad y''(0) = 1$

$\int dy = \int 2 - 6x \, dx$

$y' = 2x - \frac{\cancel{6}x^2}{\cancel{2}} + c$

$y' = 2x - 3x^2 + c$

$y'(0) = 2.0 - 3.0^2 + c = 4$

$= 0 - 3.0 + c = 4$

$= 0 - 0 + c = 4$

$c = 4$

$y' = 2x - 3x^2 + 4$

$\int dy = \int 2x - 3x^2 + 4$

$y'' = \frac{\cancel{2}x^2}{\cancel{2}} - \frac{\cancel{3}x^3}{\cancel{3}} + 4x + c$

$= x^2 - x^3 + 4x + c$

$y''(0) = 0^2 - 0^3 + 4.0 + c = 1$

$= 0 - 0 + 4.0 + c = 1$

$= 0 - 0 + 0 + c = 1$

$c = 1$

$y'' = x^2 - x^3 + 4x + 1$

f. $\frac{d^2y}{dx^2} = \frac{2}{x^3}$ $\quad y'(1) = 1 \quad y''(1) = 1$

$\int dy = \int \frac{2}{x^3} \, dx$

$y' = 2x^{-3} + c$

$y' = \frac{\cancel{2}x^{-2}}{-\cancel{2}} + c$

$y' = -x^{-2} + c$

$y'(1) = -1^{-2} + c = 1$

$= -1 + c = 1$

$c = +1 + 1$

$c = 2$

$y' = -x^{-2} + 2$

$\int dy = \int -x^{-2} + 2 \, dx$

$y'' = \frac{-x^{-1}}{-1} + 2x + c$

$y'' = x^{-1} + 2x + c$

$y''(1) = 1^{-1} + 2.1 + c = 1$

$= 1 + 2 + c = 1$

$= 3 + c = 1$

$c = 1 - 3$

$c = -2$

$y'' = x^{-1} + 2x - 2$

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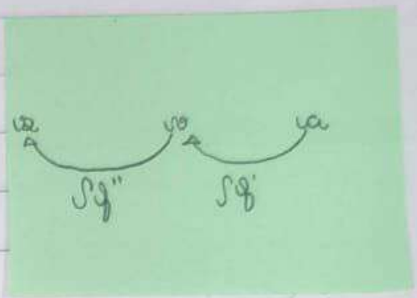
3. Uma partícula se move ao longo de uma reta com aceleração $a = 2t - 3 \, m/s^2$. Em $t = 0$ tem se $s = 0$ e ela se move com velocidade $v = 4 \, m/s$. Determine as fórmulas para a velocidade e a posição da partícula.



$a = \int 2t - 3$

$v = \frac{2t^2}{2} - 3t + c$

$v = t^2 - 3t + c$

$v(0) = 0^2 - 3 \cdot 0 + c = 4$

$= 0 + c = 4$

$= 4$

$v = t^2 - 3t + 4$

$s = \int t^2 - 3t + 4$

$s = \frac{t^3}{3} - \frac{3t^2}{2} + 4t + c$

$s(0) = \frac{0^3}{3} - \frac{3 \cdot 0^2}{2} + 4 \cdot 0 + c = 0$

$= \frac{0}{3} - \frac{0^2}{2} + 0 + c = 0$

$= 0 - 0 + 0 + c = 0$

$c = 0$

$s = \frac{t^3}{3} - \frac{3t^2}{2} + 4t + 0$

4. Uma partícula se move com velocidade dada por $v(t) = 2\cos(t) - 3t^2 \, m/s$. Sabendo que a posição inicial é $s(0) = 5 \, m$, encontre a função posição $s(t)$.

$v = \int 2\cos(t) - 3t^2$

$s = 2\operatorname{sen}(t) - \frac{3t^3}{3}$

$s = 2\operatorname{sen}(t) - t^3 + c$

$s(0) = 2\operatorname{sen}(0) - 0^3 + c = 5$

$= 2 \cdot 0 - 0 + c = 5$

$= 0 + c = 5$

$c = 5$

$s = 2\operatorname{sen}(t) - t^3 + 5$

5. Uma partícula se desloca ao longo de um eixo com aceleração $a = 15\sqrt{t}$, sujeito a $s = 4$ e $v = 0$ quando $t = 1$. Determine (a) a velocidade em termos de t e (b) a posição em termos de t.

$a = \int 15\sqrt{t} = \int 15\,t^{\frac{1}{2}}$

$v = \frac{15t^{\frac{3}{2}}}{\frac{3}{2}} = \frac{2.15t^{\frac{3}{2}}}{3} = 2.5t^{\frac{3}{2}} = 10t^{\frac{3}{2}} + c$

$v = 10t^{\frac{3}{2}} - 6$

$v(1) = 10.1^{\frac{3}{2}} + c = 4$

$= 10.1 + c = 4$

$= 10 + c = 4$

$c = 4 - 10 = -6$

$s = \int 10t^{\frac{3}{2}} - 6$

$s = \frac{10t^{\frac{5}{2}}}{\frac{5}{2}} - 6x = \frac{2.10t^{\frac{5}{2}}}{5} - 6x = \frac{20t^{\frac{5}{2}}}{5} - 6x = 4t^{\frac{5}{2}} - 6t + c$

$s = 10t^{\frac{5}{2}} - 6t + 2$

$s(1) = 4.1^{\frac{5}{2}} - 6.1 + c = 0$

$= 4.1 - 6 + c = 0$

$= 4 - 6 + c = 0$

$= -2\ c = 0$

$c = 2$

6. Um objeto move-se em linha reta com aceleração dada por $a(t) = 4t + 2\ m/s^2$. Sabendo que sua velocidade inicial é $v(0) = 2\ m/s$, encontre a velocidade quando $t = 3s$.

$a = \int 4t + 2$

$v = \frac{4t^2}{2} + 2t$

$v = 2t^2 + 2t + c$

$V(0) = 2.0^2 + 2.0 + c = 2$

$= 2.0 + 0 + c = 2$

$= 0 + c = 2$

$c = 2$

$v = 2t^2 + 2t + 2$

$v(3) = 2.3^2 + 2\,3 + 2$

$= 2.9 + 6 + 2$

$= 18 + 6 + 2$

$= 26\ s$

7. Sabendo-se que o custo marginal é $C_{mg}(x) = 0{,}08x + 3$ e que o custo fixo é de R$ 400,00, obtenha a função custo total $C(x)$.

$C = \int 0{,}08x + 3$

$C = \frac{0{,}08x^2}{2} + 3x + c$

$C = 0{,}04x^2 + 3x + c$

Custo Total $= 0{,}04x^2 + 3x + 400$

$C(0) = 0{,}04 \cdot 0^2 + 3 \cdot 0^2 + c = 400$

$= 0{,}04 \cdot 0 + 0 + c = 400$

$= 0 + 0 + c = 400$

$c = 400$

Sempre que tiver custo fixo é $C(0)$

8. O custo marginal para a produção de uma quantidade x de um bem é dado por $C_{mg}(x) = 3x^2 - 60x + 400$. Sabendo que o custo total para produzir duas unidades é de R$ 900,00, qual é o custo total para produzir 5 unidades?

$C = \int 3x^2 - 60x + 400$

$C = \frac{\cancel{3}x^3}{\cancel{3}} - \frac{\cancel{60}x^2}{\cancel{2}} + 400x + c$

$C = x^3 - 30x^2 + 400x + c$

$C(2) = 2^3 - 30 \cdot 2^2 + 400 \cdot 2 + c = 900$

$= 8 - 30 \cdot 4 + 800 + c = 900$

$= 8 - 120 + 800 + c = 900$

$= -112 + 800 + c = 900$

$= 688 + c = 900$

$c = 900 - 688$

$c = 212$

$C = x^3 - 30x^2 + 400x + 212$

$C = 5^3 - 30 \cdot 5^2 + 400 \cdot 5 + 212$

$C = 125 - 30 \cdot 25 + 2000 + 212$

$C = 125 - 750 + 2000 + 212$

$C = 2337 - 750$

$C = 1587$ reais

9. O custo marginal para se produzir x unidades de uma mercadoria é dado por $C_{mg}(x) = 5 + 10x + 2x^2$ e o custo fixo é de R$ 500,00, sabendo disto determine a função custo total e o custo total para se produzir 5 unidades.

$C = \int 5 + 10x + 2x^2$

$C = 5x + \frac{\cancel{10}x^2}{\cancel{2}} + \frac{2x^3}{3} + c$

$C = 5x + 5x^2 + \frac{2x^3}{3} + c$

$C(5) = 5.5 + 5.5^2 + \frac{2.5^3}{3} + 500$

$= 25 + 5 \cdot 25 + \frac{2 \cdot 125}{3} + 500$

$= 25 + 125 + 83,33 + 500 = 733,33$ reais

$C(0) = 5.0 + 5.0^2 + \frac{2.0^3}{3} + c = 500$

$= 0 + 5 \cdot 0 + \frac{2 \cdot 0}{3} + c = 500$

$= 0 + 0 + \frac{0}{3} + c = 500$

$= 0 + 0 + c = 500$

$= 0 + c = 500$

$c = 500$

10. Sabendo-se que o custo marginal é $C_{mg}(x) = 6x^2 - 6x + 20$ e que o custo fixo é de R$400,00, obtenha a função custo total.

$C = \int 6x^2 - 6x + 20$

$C = \frac{\cancel{6}x^3}{\cancel{3}} - \frac{\cancel{6}x^2}{\cancel{2}} + 20x + c$

$C = 2x^3 - 3x^2 + 20x + c$

$C(0) = 2.0^3 - 3.0^2 + 20.0 + c = 400$

$= 2.0 - 3.0 + 0 + c = 400$

$= 0 + c = 400$

$c = 400$

$C = 2x^3 - 3x^2 + 20x + 400$

11. Calcule as integrais indefinidas usando a substituição indicada

1a. $\int 28\underbrace{(7x-2)}_{u}^{-5}\, dx$ $\quad u = 7x - 2$

$\frac{du}{dx} = 7 = dx = \frac{du}{7} \rightarrow \int 28(u)^{-5} \frac{du}{7}$

Deriva o u

$= \frac{28}{7} \cdot \frac{u^{-4}}{-4} + c$

$= 4 \cdot \frac{u^{-4}}{-4} + c = \frac{\cancel{4}u^{-4}}{-\cancel{4}} = -u^{-4} + c$

Só substitui pelo u

"$-(7x-2)^{-4} + c$"

b. $\int \frac{9x^2}{\sqrt{1-x^3}} \, dx \qquad u = 1 - x^3$

$\frac{du}{dx} = -3x^2 \rightarrow dx = \frac{du}{-3x^2}$

$\int \frac{9x^2}{\sqrt{u}} \cdot \frac{du}{-3x^2} = \int \frac{-3}{u^{\frac{1}{2}}} \, du = -3u^{-\frac{1}{2}} \, du$

$= \frac{-3u^{\frac{1}{2}}}{\frac{1}{2}} + c$

$= \frac{2(-3)u^{\frac{1}{2}}}{1} + c$

$= -6u^{\frac{1}{2}} + c$

$u = 6(1-x^3)^{\frac{1}{2}} + c$

c. $\int \frac{e^{2\sqrt{x}}}{\sqrt{x}} \, dx \qquad u = 2\sqrt{x}$

$u = 2(x^{\frac{1}{2}})$

$\frac{du}{dx} = \frac{2 \cdot 1}{2} x^{-\frac{1}{2}} = \frac{\cancel{2}x^{-\frac{1}{2}}}{\cancel{2}} = \frac{1}{\sqrt{x}} = \sqrt{x} = \sqrt{x}\,du$

Volta o u e substitui

$\int \frac{e^{u}}{\sqrt{x}} \cdot \sqrt{x} \, du = \int \frac{e^{u}}{\sqrt{x}} \cdot \sqrt{x} \, du = \int e^{u} \, du = e^{u} + c$

$u = e^{2\sqrt{x}} + c$

d. $\int \frac{2x}{x^2+5}\,dx$ $\quad u = x^2+5$

$\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$

$\int \frac{\cancel{2x}}{u} \cdot \frac{du}{\cancel{2x}} = \frac{1du}{u} + c = \ln|u| + c$

$u = \ln|x^2+5| + c$

e. $\int \sqrt{3-2x}\,dx$ $\quad u = 3-2x$

$\frac{du}{dx} = -2 \Rightarrow dx = \frac{du}{-2}$

$\int \frac{\sqrt{u}\,du}{-2} = \int \frac{u^{\frac{1}{2}}\,du}{-2} = \frac{u^{\frac{3}{2}}}{\frac{3(-2)}{2}} + c = \frac{u^{\frac{3}{2}}}{-3} + c$

$u = \frac{3-2x^{\frac{3}{2}}}{3} + c$

f. $\int \sin(3x+4)\,dx$ $\quad u = 3x+4$ ← derivada 3

$\frac{du}{dx} = 3 \Rightarrow dx = \frac{du}{3}$

$\int \sin(u)\,\frac{du}{3} = \int \frac{\sin(u)\,du}{3} = \frac{\cos(u)}{3} + c$

$c = \frac{-\cos(3x+4)}{3} + c$

g. $\int \sin(x)\cos(x)\, dx \quad u = \sin(x)$

$\frac{du}{dx} = \cos(x) \rightarrow dx = \frac{du}{\cos(x)}$

$\int (u) \cancel{\cos(x)} \frac{du}{\cancel{\cos x}} = u + c = \frac{u^2}{2} + c$

$c = \frac{\sin x^2}{2} + c$

h. $\int \frac{(\ln x)^2}{x} dx \quad u = \ln(x)$

$\frac{du}{dx} = \frac{1}{x} \rightarrow dx = x\, du$

$\int \frac{(u)^2}{\cancel{x}} \cancel{x} du = \int u^2 du = \frac{u^3}{3} + c$

$c = \frac{(\ln(x))^3}{3} + c$

12. Calcule as integrais usando a regra da substituição

a. $\int \frac{1}{x(\ln x)^2} dx \quad u = \ln x$

$\frac{du}{dx} = \frac{1}{x} \rightarrow dx = x\, du$

$\int \frac{1}{x u^2} x\, du = \int \frac{1}{u^2} du = u^{-2} c = \frac{u^{-1}}{-1} + c = -u^{-1} + c$

$u = -\ln|x|^{-1} + c$

b. $\int \frac{1}{\sqrt{5x+4}}\,dx$ $u = 5x+4$

$\frac{du}{dx} = 5$ $dx = \frac{du}{5}$

$\int \frac{1}{\sqrt{u}}\,\frac{du}{5} = \int \frac{1}{5\sqrt{u}} = \frac{1}{5(u^{\frac{1}{2}})} + c = \frac{u^{-\frac{1}{2}}}{5} + c = \frac{u^{\frac{1}{2}} + c}{5 \cdot \frac{1}{2}} = \frac{u^{\frac{1}{2}}}{\frac{5}{2}} + c = \frac{2u^{\frac{1}{2}}}{5} + c$

$u = \frac{2(5x+4)^{\frac{1}{2}}}{5} + c$

c. $\int x^4\sqrt{1-x^2}\,dx$ $u = 1-x^2$

$\frac{du}{dx} = -2x$ $dx = \frac{du}{2x}$

$\int \cancel{x}\,\sqrt[4]{u}\,\frac{du}{-2\cancel{x}} = \int \frac{\sqrt[4]{u}\,du}{-2} = -\frac{u^{\frac{1}{4}}}{2} + c = -\frac{u^{\frac{5}{4}}}{2 \cdot \frac{5}{4}} + c = -\frac{u^{\frac{5}{4}}}{\frac{5}{2}} + c = -\frac{2u^{\frac{5}{4}}}{5}$

$u = -\frac{2(1-x^2)^{\frac{5}{4}}}{5} + c$

d. $\int x^2\left(\frac{x^3}{18} - 1\right)^5 dx$ $u = \frac{x^3}{18} - 1$

$\frac{du}{dx} = \frac{\cancel{3}x^2}{\cancel{18}} = \frac{x^2}{6}$ $dx = \frac{du}{\frac{x^2}{6}}$

$$\int x^2 (u)^5 \frac{du}{x^2} = \int \cancel{x^2} (u)^5 \frac{6\,du}{\cancel{x^2}} \int 6u^5\,du = \frac{\cancel{6}\,u^6}{\cancel{6}} + c = u^6 + c$$

$$u = \left(\frac{x^3}{18} - 1\right)^6 + c$$

e. $\int t e^{-t^2}\,dx \qquad u = -t^2$

$\frac{du}{dx} = -2t \rightarrow dx = \frac{du}{-2t}$

$$\int t e^{u} \frac{du}{-2t} = \int \frac{\cancel{t} e^{u}}{-2\cancel{t}} = \frac{e^{u}}{-2} + c$$

$$u = \frac{e^{-t^2}}{2} + c$$

f. $\int \frac{e^{\frac{1}{x}}}{x^2}\,dx \qquad u = \frac{1}{x}$

$\frac{du}{dx} = \frac{-1}{x^2} \rightarrow dx = \frac{du}{\frac{-1}{x^2}}$

$$\int \frac{e^{u}}{x^2} \frac{du}{\frac{-1}{x^2}} = \int \frac{e^{u}}{\cancel{x^2}} \frac{\cancel{x^2}\,du}{-1} = \frac{e^{u}}{-1} + c$$

$$u = e^{\frac{1}{x}} + c$$

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g. $\int \sin(x)\cos(x)\,dx$ $u = \sin(x)$

$\frac{du}{dx} = \cos(x) \to dx = \frac{du}{\cos(x)}$

$$\int (u) \cancel{\cos(x)} \frac{du}{\cancel{\cos x}} = u + c = \frac{u^2}{2} + c$$

$c = \frac{\sin x^2}{2} + c$

h. $\int \frac{(\ln x)^2}{x} dx$ $u = \ln(x)$

$\frac{du}{dx} = \frac{1}{x} \to dx = x\,du$

$$\int \frac{(u)^2}{\cancel{x}} \cancel{x}\,du = \int u^2\,du = \frac{u^3}{3} + c$$

$c = \frac{(\ln(x))^3}{3} + c$

12. Calcule as integrais usando a regra da substituição

a. $\int \frac{1}{x(\ln x)^2} dx$ $u = \ln x$

$\frac{du}{dx} = \frac{1}{x} \to dx = x\,du$

$$\int \frac{1}{xu^2} x\,du = \int \frac{1}{u^2} du = u^{-2} c = \frac{u^{-1}}{-1} + c = -u^{-1} + c$$

$u = -\ln|x|^{-1} + c$

b. $\int \frac{1}{\sqrt{5x+4}} \, dx \qquad u = 5x+4$

$\frac{du}{dx} = 5 \qquad dx = \frac{du}{5}$

$\int \frac{1}{\sqrt{u}} \, \frac{du}{5} = \int \frac{1}{5\sqrt{u}} = \frac{1}{5(u^{\frac{1}{2}})} + c = \frac{u^{-\frac{1}{2}}}{5} + c = \frac{u^{\frac{1}{2}}}{5 \cdot \frac{1}{2}} + c = \frac{u^{\frac{1}{2}}}{\frac{5}{2}} + c = \frac{2u^{\frac{1}{2}}}{5} + c$

$u = \frac{2(5x+4)^{\frac{1}{2}}}{5} + c$

c. $\int x^4 \sqrt{1-x^2} \, dx \qquad u = 1-x^2$

$\frac{du}{dx} = -2x \qquad dx = \frac{du}{2x}$

$\int x^4 \sqrt{u} \, \frac{du}{-2\cancel{x}} = \int \frac{\sqrt[4]{u} \, du}{-2} = -\frac{u^{\frac{1}{4}}}{2} + c = -\frac{u^{\frac{5}{4}}}{2 \cdot \frac{5}{4}} + c = -\frac{u^{\frac{5}{4}}}{\frac{5}{2}} + c = -\frac{2u^{\frac{5}{4}}}{5}$

$u = -\frac{2(1-x^2)^{\frac{5}{4}}}{5} + c$

d. $\int x^2 \left(\frac{x^3}{18} - 1\right)^5 dx \qquad u = \frac{x^3}{18} - 1$

$\frac{du}{dx} = \frac{\cancel{3}x^2}{\cancel{18}} = \frac{x^2}{6} \qquad dx = \frac{du}{\frac{x^2}{6}}$

$$\int x^2 (u)^5 \frac{du}{x^2} = \int \cancel{x^2} (u)^5 \frac{6du}{\cancel{x^2}} \int 6u^5 du = \frac{\cancel{6} u^6}{\cancel{6}} + c = u^6 + c$$

$$u = \left(\frac{x^3}{18} - 1\right)^6 + c$$

12. $\int t e^{-t^2} dx \qquad u = -t^2$

$\frac{du}{dx} = -2t \rightarrow dx = \frac{du}{-2t}$

$$\int t e^u \frac{du}{-2t} = \int \frac{\cancel{t} e^u}{-2\cancel{t}} = \frac{e^u}{-2} + c$$

$$u = \frac{e^{-t^2}}{2} + c$$

13. $\int \frac{e^{\frac{1}{x}}}{x^2} dx \qquad u = \frac{1}{x}$

$\frac{du}{dx} = \frac{-1}{x^2} \rightarrow dx = \frac{du}{\frac{-1}{x^2}}$

$$\int \frac{e^u}{x^2} \frac{du}{\frac{-1}{x^2}} = \int \frac{e^u}{\cancel{x^2}} \frac{\cancel{x^2} du}{-1} = \frac{e^u}{-1} + c$$

$$u = e^{\frac{1}{x}} + c$$

g. $\int e^x \sqrt{e^x - 2}\, dx \qquad u = e^x - 2$

$\frac{du}{dx} = e^x \rightarrow dx = \frac{du}{e^x}$

$$\int e^x \sqrt{u}\, \frac{du}{e^x} = \int u^{\frac{1}{2}}\, du = \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{2u^{\frac{3}{2}}}{3} + c$$

$$= \frac{2(e^x - 2)^{\frac{3}{2}}}{3} + c$$

h. $\int \frac{e^x}{1+e^x}\, dx \qquad u = 1 + e^x$

$\frac{du}{dx} = e^x \rightarrow dx = \frac{du}{e^x}$

$$\int \frac{\cancel{e^x}}{u}\, \frac{du}{\cancel{e^x}} = \int \frac{1}{u}\, du = \ln|u| + c$$

derivada de $\ln(x) = \frac{1}{x}$

$u = \ln|1 + e^x| + c$

i. $\int \frac{1}{x\sqrt{\ln(x)+1}}\, dx \qquad u = \ln(x) + 1$

$\frac{du}{dx} = \frac{1}{x} \rightarrow dx = \frac{du}{\frac{1}{x}}$

$$\int \frac{1}{x\sqrt{u}}\, \frac{du}{\frac{1}{x}} = \int \frac{1}{\cancel{x}\sqrt{u}}\, \frac{\cancel{x}\, du}{1} = \frac{1}{\sqrt{u}} + c = \frac{1}{u^{\frac{1}{2}}} + c = u^{-\frac{1}{2}} + c = \frac{u^{\frac{1}{2}}}{\frac{1}{2}} + c = 2u^{\frac{1}{2}} + c$$

$$= 2\sqrt{u} + c$$

$u = 2\sqrt{(\ln|x|+4)} + c$

7- $\int x \, sen(x^2) \, dx$ $\quad u = x^2$

$\frac{du}{dx} = 2x \rightarrow dx = \frac{du}{2x}$

$$\int x \, sen(u) \frac{du}{2x} = \int \frac{sen(u)}{2} du = \frac{-cos(u)}{2} + c$$

$u = \frac{-cos(x^2)}{2} + c$

8- $\int (cos(x)) e^{sen(x)} dx$ $\quad u = sen(x)$

$\frac{du}{dx} = cos(x) \rightarrow \frac{du}{cos(x)}$

Derivada sen = cos
cos = - sen

Integral sen = - cos
cos = sen

$$\int (\cancel{cos}(x)) e^{u} \frac{du}{\cancel{cos}(x)} = \int e^{u} = e^{u} + c$$

$u = e^{sen(x)} + c$

9. $\int \frac{cos(\ln(x))}{x} dx$ $\quad u = \ln(x)$

Numa fração quando não tem letra em cima ou embaixo só em cima

$\frac{du}{dx} = \frac{1}{x} \rightarrow \frac{du}{\frac{1}{x}}$

$$\int \frac{cos(u)}{x} \frac{du}{\frac{1}{x}} = \int \frac{cos(u)}{\cancel{x}} \frac{\cancel{x} du}{1} = \int \frac{cos(u)}{1} = sen(u) + c$$

$u = sen(\ln|x|) + c$

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13. Calcule as derivadas definidas

1a. $\int_0^1 (3-x)\,dx = \left[3x - \frac{x^2}{2}\right]_0^1 = \left(3 - \frac{1}{2}\right) - \left(0 - \frac{0}{2}\right)$

$= \frac{6-1}{2} - \frac{0-0}{2} \Rightarrow \frac{5}{2} - \frac{0}{2} = \frac{5}{2}$

1b. $\int_0^2 e^x\,dx = \left[e^x\right]_0^2 = 7,3890 - 1 = 6,389$ ou $e^2 - 1$

1c. $\int_{-1}^3 (3x^2-2x+1)\,dx = \left[\frac{\cancel{3}x^3}{\cancel{3}} - \frac{\cancel{2}x^2}{\cancel{2}} + x\right]_{-1}^{3} = \left[x^3-x^2+x\right]_{-1}^3 = (27-9+3)-(-1-1-1)$

$= 21-(-3) = 24$

1d. $\int_1^{32} x^{-\frac{6}{5}}\,dx = \left[\frac{x^{-\frac{1}{5}}}{-\frac{1}{5}}\right]_1^{32} = \left[\frac{5x^{-\frac{1}{5}}}{-1}\right]_1^{32} = \left[-5x^{-\frac{1}{5}}\right]_1^{32} = \left[\frac{-5}{\sqrt[5]{x}}\right]_1^{32} = \left(\frac{-5}{\sqrt[5]{32}}\right) - \left(\frac{-5}{\sqrt[5]{1}}\right)$

$= \left(\frac{-5}{2}\right) - \left(\frac{-5}{1}\right)$

$= \frac{-5-(-10)}{2}$

$= \frac{-5+10}{2} = \frac{5}{2}$

1e. $\int_0^1 (x^2+\sqrt{x})\,dx = \left[\frac{x^3}{3} + \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^1 = \left[\frac{x^3}{3} + \frac{2x^{\frac{3}{2}}}{3}\right]_0^1 = \left(\frac{1}{3} + \frac{2}{3}\right) - \left(\frac{0}{3} + \frac{0}{3}\right)$

$= \frac{3}{3} - \frac{0}{3} = \frac{3}{3}$

1f. $\int_{\sqrt{2}}^1 \left(\frac{u^7}{2} - \frac{1}{u^5}\right)du = \int_{\sqrt{2}}^1 \left(\frac{u^7}{2} - \frac{u^{-5}}{1}\right) = \left[\frac{u^8}{28} - \frac{u^{-4}}{-4}\right]_{\sqrt{2}}^1 = \left(\frac{u^8}{16} + \frac{u^{-4}}{4}\right)_{\sqrt{2}}^1$

$=$

$$\left(\frac{1}{16}-\frac{1}{4}\right)-\left(\frac{(\sqrt{2})^{8}}{16}+\frac{(\sqrt{2})^{-4}}{4}\right)=\left(\frac{5}{16}\right)-\left(\frac{1+\frac{1}{4}}{4}\right)=\frac{5}{16}-\frac{1}{8}-\frac{1}{16}=\frac{5-16-2}{16}=\frac{-13}{16}$$

g- $\int_{-1}^{1} \frac{5x}{(4+x^2)^2}\,dx =$ $u=4+x^2$ $\left(\frac{-5}{10}\right)+\left(\frac{-5}{10}\right) = \frac{-5}{10}+\frac{5}{10}=0$

$\frac{du}{dx}=2x \to dx=\frac{du}{2x}$

$$\int_{-1}^{1}\frac{5x}{u^2}\,\frac{du}{2x}=\int_{-1}^{1}\frac{5}{u^2}\,du=\left(\frac{5u^{-2}}{2}\right)_{-1}^{1}=\left(\frac{5u^{-1}}{2\cdot -1}\right)_{-1}^{1}=\left(\frac{-5(4+x^2)^{-1}}{-2}\right)_{-1}^{1}=\frac{-5}{2(4+x^2)}$$

h- $\int_0^1 \sqrt{t^5+2t}\,(5t^4+2)\,dt =$ $u=t^5+2t$ $\frac{2(3)^{\frac{3}{2}}}{3}-0=\frac{2(3)^{\frac{3}{2}}}{3}$

$\frac{du}{dt}=5t^4+2 \to dt=\frac{du}{5t^4+2}$

$$\int_0^1 \sqrt{u}\,(\cancel{5t^4+2})\,\frac{du}{\cancel{5t^4+2}}=\left(u^{\frac{1}{2}}\right)_0^1=\left(\frac{u^{\frac{3}{2}}}{\frac{3}{2}}\right)_0^1=\left(\frac{2u^{\frac{3}{2}}}{3}\right)_0^1=\left(\frac{2(t^5+2t)^{\frac{3}{2}}}{3}\right)_0^1$$

i- $\int_1^2 \frac{2\ln(x)}{x}\,dx =$ $u=\ln(x)$

j- $\int_0^{\pi} \sin(x)\,dx = [-\cos(x)]_0^{\pi}$ $=-(-1)-(-1)$

$=1+1$

$=2$

14- Determine a área da região compreendida entre as curvas e as retas

a. $y=x^2-2$ e $y=2$

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$\begin{cases} y = x^2 - 2 \\ y = 2 \end{cases}$ $\rightarrow x^2 - 2 = 2$ $x^2 - 4 = 0$

$x^2 - 2 - 2 = 0$ $x^2 = 4$

$x^2 - 4 = 0$ $\pm\sqrt{4} \rightarrow +2 \; -2$

Logo $x' = 2$

$x'' = -2$

$$\int_{-2}^{2} 2 - (x^2 - 2)\, dx = \int_{-2}^{2} (2 - x^2 - 2)\, dx = \int_{-2}^{2} 4 - x^2\, dx = \left[4x - \frac{x^3}{3} \right]_{-2}^{2}$$

$$= \left(8 - \frac{8}{3}\right) - \left(-8 - \left(\frac{-8}{3}\right)\right)$$

$$= \frac{24 - 8}{3} - \left(-8 + \frac{8}{3}\right)$$

$$= \frac{16}{3} - \frac{-24 + 8}{3} = \frac{16}{3} - \left(\frac{-16}{3}\right)$$

$$= \frac{32}{3} \text{ u.a}$$

12. $y = x^2$ e $y = 2x$

$\begin{cases} y = x^2 \\ y = 2x \end{cases}$ $\rightarrow x^2 = 2x$

$x^2 - 2x = 0$

$a = 1 \quad b = -2 \quad c = 0$

$\Delta = -2^2 - 4 \cdot 1 \cdot 0$

$\Delta = 4 - 0$

$\Delta = 4$

$x = \frac{-(-2) \pm \sqrt{4}}{2 \cdot 1} = \frac{2 \pm 2}{2}$

$x' = \frac{2 + 2}{2} = \frac{4}{2} = 2$

$x'' = \frac{2 - 2}{2} = \frac{0}{2} = 0$

Logo $x' = 2$

$x'' = 0$

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$$\int_0^2 x^2-2x\,dx = \left(\frac{x^3}{3} - \frac{2x^2}{2}\right)_0^2 = \left(\frac{x^3}{3} - x^2\right)_0^2 = \left(\frac{8}{3} - 4\right) - \left(\frac{0}{3} - 0\right)$$

$$= \frac{8-12}{3} - \frac{0}{3}$$

* Use módulo para transformar em positivo

$$= -\frac{4}{3} = \frac{4}{3}\ u.a$$

c. $y = x^3$ e $y = x$

$\begin{cases} y = x^3 \\ y = x \end{cases} \Rightarrow x^3 = x$

$x^3 - x = 0$

$x(x^2-1) = 0$

$a = 1 \quad b = 0 \quad c = -1$

$\Delta = 0^2 - 4.1.-1$

$\Delta = 4$

$x = \frac{-0 \pm \sqrt{4}}{2.1} = \frac{0 \pm 2}{2}$

$x' = \frac{0+2}{2} = \frac{2}{2} = 1$

$x'' = \frac{0-2}{2} = \frac{-2}{2} = -1$

Logo $x = 0$

$x' = 1$

$x' = -1$

$$\int_1^0 x^3-x = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)_1^0 = \left(\frac{0}{4} - \frac{0}{2}\right) - \left(\frac{1}{4} - \frac{1}{2}\right)$$

$$\left(\frac{0-0}{4}\right) - \left(\frac{1-2}{4}\right) = \frac{0}{4} - \left(\frac{-1}{4}\right) = \frac{1}{4}$$

$$\int_{-1}^0 x^3-x = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)_{-1}^0 = \left(\frac{0}{4} - \frac{0}{2}\right) - \left(\frac{1}{4} - \frac{1}{2}\right)$$

$$= \left(\frac{0-0}{4}\right) - \left(\frac{1-2}{4}\right) = \frac{0}{4} - -\frac{1}{4} = \frac{1}{4}$$

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$\frac{1}{4} + \frac{1}{4} = \frac{\cancel{2}^{3}}{\cancel{4}^{2}} = \frac{1}{2}$ u.a

1d. $y = x^4$ e $y = 8x$

$\begin{cases} y = x^4 \\ y = 8x \end{cases}$ $\Rightarrow x^4 = 8x$

$x^4 - 8x = 0$

$x(x^3 - 8)$

$x^3 - 8 = 0$

$x^3 = 8$

$x = \sqrt[3]{8} = 2$

Logo $x = 0$

$x = 2$

$$\int_2^0 (x^4 - 8x)\,dx = \left(\frac{x^5}{5} - \frac{8x^2}{2}\right)_2^0 = \left(\frac{x^5}{5} - 4x^2\right)_2^0 = \left(\frac{0}{5} - 0\right) - \left(\frac{32}{5} - 16\right)$$

$$= \frac{0}{5} - \left(\frac{32 - 80}{5}\right)$$

$$= -\left(\frac{-48}{5}\right) = \frac{48}{5} \text{ u.a}$$

* Usa módulo para transformar em positivo

1e. $y = x^2$ e $y = -x^2 + 4x$

$\begin{cases} y = x^2 \\ y = -x^2 + 4x \end{cases}$ $\Rightarrow x^2 = -x^2 + 4x$

$x^2 + x^2 - 4x = 0$

$2x^2 - 4x = 0$

$a = 2 \quad b = -4 \quad c = 0$

$\Delta = -4^2 - 4.2.0$

$\Delta = 16 - 0$

$x = \frac{-(-4) \pm \sqrt{16}}{2.2} = \frac{4 \pm 4}{4}$

$x' = \frac{4 + 4}{4} = \frac{8}{4} = 2$

$x'' = \frac{4 - 4}{4} = \frac{0}{4} = 0$

Logo $x = 2$

$x = 0$

$$\int_0^2 x^2-(-x^2+4x)\,dx = \left(\frac{x^3}{3} - \left(\frac{-x^3}{3} + \frac{4x^2}{2}\right)\right)\Big|_0^2$$

$$= \left(\frac{8}{3} - \left(\frac{-8}{3} + \frac{8}{2}\right)\right)$$

$$= \frac{8}{3} - \left(\frac{-16+24}{6}\right) = \frac{8}{3} - \frac{8}{6} = \frac{16-8}{6} = \frac{8}{3} \text{ u.a}$$

f- $y = x^2$ e $y = x^3$

$$\begin{cases} y = x^2 \\ y = x^3 \end{cases} \Rightarrow x^2 = x^3$$

$$x^2 - x^3 = 0$$

$$x(x-x^2) = 0$$

$a = -1 \quad b = 1 \quad c = 0$

$\Delta = 1^2 - 4.-1.0$

$\Delta = 1$

Logo $x = 0$

$x = 1$

$$\int_1^0 (x^2-x^3) = \left(\frac{x^3}{3} - \frac{x^4}{4}\right)\Big|_1^0 = \left(\frac{1}{3} - \frac{1}{4}\right) = \frac{4-3}{12} = \frac{1}{12} \text{ u.a}$$

g- $y = x^2 - x$ e $y = x - x^2$

$$\begin{cases} y = x^2 - x \\ y = x - x^2 \end{cases} \Rightarrow x^2 - x = x - x^2$$

$$x^2 - x - x + x^2 = 0$$

$$2x^2 - 2x$$

$a = 2 \quad b = -2 \quad c = 0$

$\Delta = -2^2 - 4.2.0$

$\Delta = 4 - 0$

$\Delta = 4$

$x = \frac{-(-2) \pm \sqrt{4}}{2.2} = \frac{2 \pm 2}{4}$

$x' = \frac{2+2}{4} = \frac{4}{4} = 1$

$x'' = \frac{2-2}{4} = \frac{0}{4} = 0$

Logo $x = 1$

$x = 0$

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$$\int_0^1 (x^2-x)-(x-x^2)\,dx = \left(\left(\frac{x^3}{3}-\frac{x^2}{2}\right)-\left(\frac{x^2}{2}-\frac{x^3}{3}\right)\right)\Big|_0^1$$

$$= \left(\frac{1}{3}-\frac{1}{2}\right)-\left(\frac{1}{2}-\frac{1}{3}\right)=\left(\frac{2-3}{6}\right)-\left(\frac{3-2}{6}\right)$$

$$= -\frac{1}{6}-\frac{1}{6} = \frac{\cancel{2}}{\cancel{6}} = \frac{1}{3} \text{ u.a}$$

* b. $y=\sqrt{x}$ e $y=x^2$

$$\begin{cases} y=\sqrt{x} \\ y=x^2 \end{cases}$$

$\to \sqrt{x}=x^2$

$(\sqrt{x})^2=x^4$

$x=x^4$

$x^4-x=0$

$x(x^3-1)=0$

$x^3-1=0$

$x^3=1 \to x=1$

Logo $x=0$

$x=1$

$$\int_1^0 x^{\frac{1}{2}}-x^2\,dx = \left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}-\frac{x^3}{3}\right)\Big|_1^0 = \left(\frac{2x^{\frac{3}{2}}}{3}-\frac{x^3}{3}\right)\Big|_1^0 = \frac{2}{3}-\frac{1}{3}=\frac{1}{3} \text{ u.a}$$

c. $x=-y^2$ e $x=y-6$

$$\begin{cases} x=-y^2 \\ x=y-6 \end{cases}$$

$\to -y^2=y-6$

$-y^2-y+6=0$

$a=-1 \quad b=-1 \quad c=6$

$\Delta=-1^2-4\cdot-1\cdot6$

$\Delta=1+24$

$\Delta=25$

$x=\frac{-(-1)\pm\sqrt{25}}{2\cdot-1}=\frac{1\pm5}{-2}$

$x'=\frac{1+5}{-2}=\frac{6}{-2}=3$

$x''=\frac{1-5}{-2}=\frac{-4}{-2}=2$

Logo $x=2$

$x=-3$

$$\int_{2}^{-3} -y^2 - y + 6 = \left(\frac{-y^3}{3} - \frac{y^2}{2} + 6y\right)_2^{-3} = \left(\frac{-8}{3} - 2 + 12\right) - \left(\frac{9}{2} - 9 - 18\right)$$

$$= \left(\frac{-8 - 6 + 36}{3}\right) - \left(18 - 9 - 36\right)$$

$$= \frac{22}{3} - \left(\frac{-21}{2}\right) = \frac{44 + 81}{6} = \frac{125}{6} \text{ u a}$$

$y - x + y^2 = 0$ e $x + 3y^2 = 2$

$$\begin{cases} x + y^2 = 0 \\ x + 3y = 2 \end{cases} \rightarrow x = -y^2 \quad x = -3y + 2 \quad \rightarrow -y^2 = -3y + 2 \quad 2y^3 - 2 = 0$$

Logo $x = -1$
$x = 1$

$$\int_{-1}^{1} -3y^2 + 2 + y^2 \, dy = \int \frac{-3y^3}{3} + 2u + \frac{y^3}{3} = \int \frac{-2u^3}{3} + 2y$$

$$\left(\frac{-2}{3} + 2\right) - \left(\frac{2}{3} - 2\right) = -\frac{2}{3} + 2 - \frac{2}{3} + 2$$

$$= \frac{-2 - 2 + 6 + 6}{3} = \frac{8}{3} \text{ u a}$$

15. Determine a área da região entre a curva e o eixo X.

a. $y = -x^2 - 2x, \ -2 \leq x \leq 2$

$-x^2 - 2x = 0$ Logo $x = 0$
$-x(x + 2) = 0$ $x = -2$
$x = 2$

$$\int_{-2}^{0} -x^2 - 2x = \left(\frac{-x^3}{3} - \frac{\cancel{2}x^2}{\cancel{2}} \right)_{-2}^{0} = \left(\frac{-x^3}{3} - x^2 \right)_{-2}^{0}$$

$$= \left(\frac{8}{3} + 4 \right) = \frac{8+12}{3} = \frac{20}{3}$$

$$\int_{2}^{0} -x^2 - 2x = \left(\frac{-x^3}{3} - \frac{\cancel{2}x^2}{\cancel{2}} \right)_{2}^{0} = \left(\frac{-x^3}{3} - x^2 \right)_{0}^{2}$$

$$* = \left(\frac{-8}{3} + 4 \right) = \frac{-8+12}{3} = \frac{4}{3}$$

$$\frac{20}{3} + \frac{4}{3} = \frac{24}{3}$$

2o. $y = x^3, \; -1 < x < 4$

$x^3 = 0$ Logo $x = 0$

$x(x^2) = 0$ $x = -1$

$x = 4$

$$\int_{-1}^{0} x^3 \, dx = \left(\frac{x^4}{4} \right)_{-1}^{0} = \frac{1}{4}$$

$$\int_{4}^{0} x^3 \, dx = \left(\frac{x^4}{4} \right)_{4}^{0} = \frac{256}{4}$$

$$\frac{256}{4} + \frac{1}{4} = \frac{257}{4} \text{ u.a.}$$

16. Determine a área da região em forma de hélice entre a curva $X - y^3 = 0$ e a reta $X - y = 0$.

$$\begin{cases} X = y^3 \\ X = y \end{cases} \Rightarrow y^3 = y \qquad y^3 - y = 0 \qquad y(y^2 - 1) = 0 \qquad y^2 + 1 = 0 \qquad y' = 1 \qquad y'' = -1$$

Ficou positivo porque usei o módulo

Logo $X = 0$
$X = -1$
$X = 1$

$$\frac{-1}{4} + \frac{-1}{4} = \frac{-2^2}{4^2} = \frac{-1}{2}$$

$$= \frac{1}{2} \text{ u.a}$$

$$\int_1^0 y^3 - y \, dy = \left(\frac{y^4}{4} - \frac{y^2}{2} \right)\Big|_1^0 = \frac{1}{4} - \frac{1}{2} = \frac{1-2}{4} = \frac{-1}{4}$$

$$\int_{-1}^0 y^3 - y \, dy = \left(\frac{y^4}{4} - \frac{y^2}{2} \right)\Big|_{-1}^0 = \frac{1}{4} - \frac{1}{2} = \frac{1-2}{4} = \frac{-1}{4}$$

17. Determine a área da região no primeiro quadrante, limitada acima pela curva $y = e^{2X}$, abaixo pela curva $y = e^x$ e à direita pela reta $X = \ln 3$.

$$\begin{cases} y = e^X \\ y = e^{2X} \end{cases} \Rightarrow e^x = e^{2x}$$

$$\ln(e^x) = \ln(e^{2x})$$

$$X = 2X$$

$$-2X + X = 0$$

$$-X = 0$$

$$X = 0$$

Logo $X = \ln 3$
$X = 0$

$$\int_0^{\ln 3} e^x - e^{2x} \, dx = \left(e^x - \frac{e^{2x}}{2} \right)\Big|_0^{\ln 3} = \left(e^{\ln 3} - \frac{e^{2\ln 3}}{2} \right) - \left(e^0 - \frac{e^{20}}{2} \right)$$

$$= \left(3 - \frac{9}{2} \right) - \left(1 + \frac{1}{2} \right) = \frac{6 - 9 - 2 + 1}{2} = 2$$

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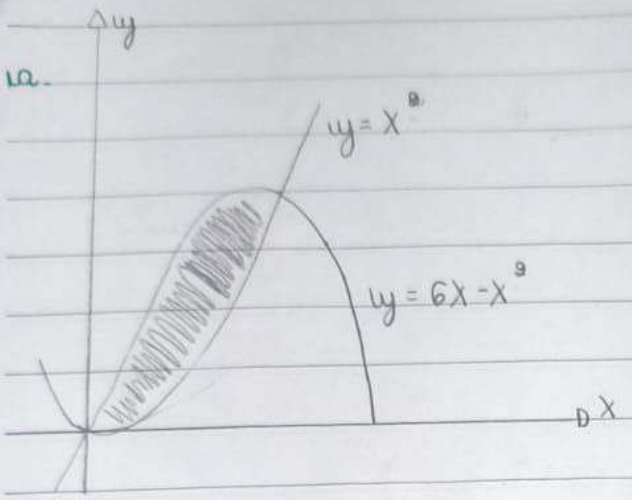
18. Calcule a área das regiões sombreadas:

a.



$$\begin{cases} y = x^2 \\ y = 6x - x^2 \end{cases} \rightarrow y = x^2 = 6x - x^2$$

$$x^2 - 6x + x^2 = 0$$

$$2x^2 - 6x = 0$$

$a = 2 \quad b = -6 \quad c = 0$

$\Delta = -6^2 - 4.2.0$

$\Delta = 36 - 8$

$\Delta = 36$

$x = \dfrac{-(-6) \pm \sqrt{36}}{2.2} = \dfrac{+6 \pm 6}{4}$

Logo $x = 0$

$x = 3$

$x' = \dfrac{6+6}{4} = \dfrac{12}{4} = 3$

$x'' = \dfrac{6-6}{4} = \dfrac{0}{4} = 0$

$$\int_3^0 x^2 - 6x - x^2 = \left(\frac{x^3}{3} - \frac{\cancel{6}x^2}{\cancel{2}} - \frac{x^3}{3}\right)\Big|_3^0 = \left(\frac{x^3}{3} - 3x^2 - \frac{x^3}{3}\right)$$

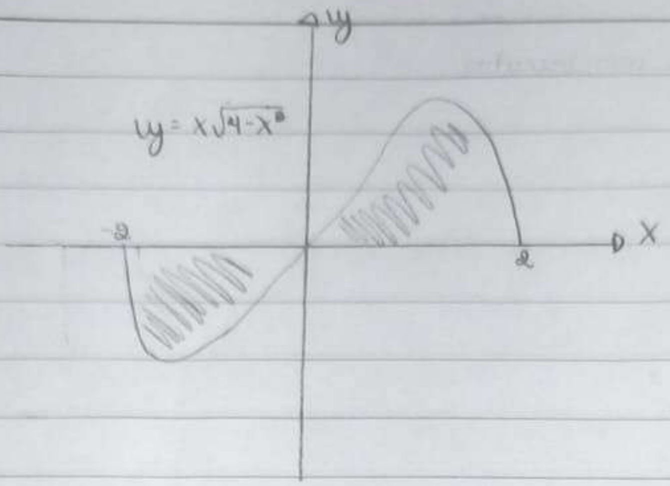
$$= \left(\frac{27}{3} - 27 - \frac{27}{3}\right) = \frac{27 - 81 - 27}{3}$$

$$= \frac{27 - 108}{3} = \frac{-81}{3} = -27 \text{ u.a}$$

$$= 27 \text{ u.a}$$

ficou positivo porque usei o módulo

b.



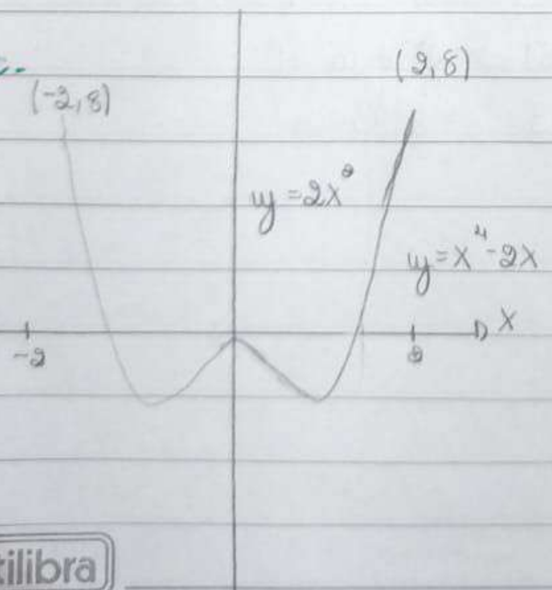
$$\int_{2}^{-2} x\sqrt{4-x^2}\,dx \qquad u = 4 - x^2$$

$du = -2x \Rightarrow dx = \frac{du}{-2x}$
dx

$$\int_{-2}^{2} x\sqrt{u}\,\frac{du}{-2x} = \left(\frac{1}{-2}u^{\frac{1}{2}}\,du\right)_{-2}^{2} = \left(\frac{1u^{\frac{3}{2}}}{-\cancel{2}\cdot 3 \cdot \frac{}{\cancel{2}}}\right)_{-2}^{2} = \frac{-1}{3}u^{\frac{3}{2}} = \frac{-8}{3}\ u.a$$

$A = 2 \cdot \frac{8}{3} = \frac{16}{3}\ u.a$

c.



$$\begin{cases} y = 2x^2 \\ y = x^4 - 2x^2 \end{cases} \rightarrow \begin{array}{l} 2x^2 = x^4 - 2x^2 \\ 2x - x^4 + 2x^2 = 0 \end{array}$$

$$\int_0^{(2)(-2)} 2x^2 - x^4 + 2x^2 = \left(\frac{2x^3}{3} - \frac{x^5}{5} + \frac{2x^3}{3} \right) = \left(\frac{2x^3}{3} - \frac{x^5}{5} + \frac{2x^3}{3} \right)_0^{(2)(-2)}$$

$$= \left(\frac{16}{3} - \frac{32}{5} + \frac{16}{3} \right) + \left(\frac{-16}{3} + \frac{32}{5} - \frac{16}{3} \right)$$

$$= \left(\frac{80 - 96 + 80}{15} \right) + \left(\frac{-80 + 96 - 80}{15} \right)$$

$$= \frac{64}{15} + \frac{64}{15} = \frac{128}{15} \text{ u.a}$$